aquasec

RUNTIME SECURITY VALIDATION

Aqua Probe

User Guide

The Aqua Security Runtime Input Validator Tool

Kubernetes workloads and Linux or Windows hosts.
Guided testing for workshops, Compass classes, demos, and proofs of value.



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Maintained fork. Use only in authorized, disposable lab environments.

01 / ORIENTATION

A guide to repeatable validation

Aqua Probe connects policy setup, runtime activity, and evidence review. Use the interactive CLI to select a control, prepare its policy, exercise a signature, and inspect the result in Aqua.

The operator configures policies. The script does not automatically create or restore Aqua policies, and this repository does not establish official Aqua product support.

In this guide	Page
Safety and scope	3
Installation and first run	4
Current terminal menu	5
Container and workload controls	6
Test activity and evidence	7
SSH host setup	8
Host controls and authentication	9
Windows registry workflows	10
Secure AI	11
CLI and image configuration	12
Cleanup and troubleshooting	13
References and maintenance	14

02 / BEFORE YOU BEGIN

Safety and scope

A disposable lab is required

Tests execute real commands and create real resources. They can deploy privileged pods, create cluster-scoped RBAC, contact external destinations, scan ports, and generate failed logins. These actions may trigger detections or affect the environment.

Reserve aqua-probe-lab exclusively

Kubernetes test workloads use **aqua-probe-lab**. The namespace does not isolate cluster-scoped tests. Main-menu termination deletes the entire namespace and named cluster roles/bindings. Do not share those names with unrelated resources.

Treat high-impact tests separately

Container File Integrity Monitoring changes permissions and deletes **/etc/sudoers** inside the disposable test container. System Integrity Monitoring attempts to set the clock to 10:00:00. With sufficient privileges, a clock change can affect the host. Never use production targets.

Keep recovery access

For authentication-policy tests, retain an independent administrator session and out-of-band console access. Use a dedicated account and check lockout thresholds. Restore policy changes manually after testing or cancellation.

Distinguish policy scopes

Container controls apply to workloads. Host Runtime Policies apply to native host activity. Windows registry tests require Windows Aqua Enforcer. Shared commands can exercise similar signatures, but they do not make these scopes interchangeable.

Evidence, not assumptions

A command failure alone does not prove an Aqua block. Check target, policy, time, and event details. Audit records remain after cleanup.

03 / QUICK START

Installation and first run

Local requirements

Use Bash and Git. Kubernetes workflows also need a configured kubectl context, registry access, and permissions for the selected tests and cleanup. Confirm the Aqua Enforcer DaemonSet and aqua-kube-enforcer deployment are present.

```
git clone https://github.com/ericgomes56/aqua-probe.git
cd aqua-probe
chmod +x aqua-probe.sh
./aqua-probe.sh --help
```

Kubernetes workflow

```
kubectl config current-context
./aqua-probe.sh
```

Confirm the context points to your lab. Startup checks Kubernetes connectivity and resource existence, not Enforcer readiness or policy health. Choose **1. Deploy/Redeploy test container**, then a control.

Host-only workflow

```
./aqua-probe.sh --host-runtime
```

This mode skips Kubernetes startup checks. Select/change SSH host, choose the OS, and provide the connection details. Each host test still requires confirmation.

Default test image

ericgomes56/aqua-probe:2.0. Custom images must provide the tools required by the control. Read prerequisites before running a test.

04 / INTERFACE

Security controls menu



Captured November 2, 2026. Main-menu numbering differs from host-only mode.

05 / COVERAGE

Container and workload controls

The main menu groups workload tests separately from host controls. Prepare the specific Runtime or Assurance Policy requested by each test.

Main menu	Controls
3-9	Bad DNS/IP Reputation; Block Container Exec; Block Cryptocurrency Mining; Block Fileless Execution; Block Non-compliant Images; Block Non-compliant Resources; Block Non-Kubernetes Containers
10-16	Block Outdated Images; Block Reverse Shell; Block Unregistered Images; Capabilities Block; Drift Prevention; Executables Allowed; Executables Blocked
17-24	File Block; File Integrity Monitoring; Forensics; Limit Container Privileges; Limit New Privileges; Package Block; Port Block; Port Scanning Detection
25-31	Real-time Malware Protection; Registries Allowed; Secure AI - Discovery; Secure AI - Protection; Secure AI - Unauthorized Models; System Integrity Monitoring; Volumes Blocked

Admission and assurance tests

For non-compliant images, configure malware checks in the Assurance Policy. For unregistered images, block unregistered images and disable automatic registration of running containers/discovered pod images as applicable. Follow the script's exact prompts.

Coverage boundaries

Limit Container Privileges automates seven Linux pod cases. Host UTS and Windows low-port cases remain manual. Capabilities Block uses CHOWN as its focused test. A default image is not guaranteed to trigger outdated-image or registry restrictions.

Full control reference and policy instructions

06 / VALIDATION

Test activity and evidence

The script displays test activity and executes it against the selected workload. Use the menu so namespace and container targeting follow the current implementation.

Control	Activity	What to verify
Drift Prevention	Copy /bin/wget to /tmp/wget_copy, then run the copy against google.com.	New executable activity and matching Aqua evidence.
Cryptocurrency Mining	wget http://crypto-loot.com	Reputation/signature event for the expected target.
DNS/IP Reputation	curl http://68.183.212.246	Network event matched to policy and time.
File Block	cat /etc/passwd	A denial attributable to the configured file rule.
Package Block	tar	Aqua evidence, not tar's ordinary missing-arguments error.
Forensics	Separate echo, id, and ls operations.	Collected process/argument activity. A block is not required.

Review in Aqua

Open **Security Reports -> Audit** and match the target, policy, timestamp, and command/resource. Use the relevant forensic or AI view for collection/discovery tests. Save sanitized output and the corresponding event for the workshop record.

Do not misclassify infrastructure errors

Missing binaries, container-not-found errors, image-pull failures, Kubernetes admission denials, and SSH failures are not proof of Aqua enforcement.

07 / HOST RUNTIME

SSH host setup

Select the target OS

Main option **2** or host-only option **13** selects/changes the SSH host. Enter Linux or Windows, hostname/IP, username, port, and an optional local private-key path such as ~/.ssh/test.pem. Enter paths without surrounding quotes.

Linux and Windows connections are retained separately for the current session. Controls reuse the required OS connection or prompt for one. SSH host-key verification remains enabled. The script does not automatically install packages, elevate privileges, or enable SSH.

Target	Requirements
Linux	Linux Aqua Enforcer, Bash, SSH, and utilities used by the chosen test. Host Secure AI also requires native Python 3 and outbound HTTPS.
Windows	Windows Aqua Enforcer, Windows OpenSSH Server, and Windows PowerShell 5.1+ available as powershell.exe. Registry tests also require local ssh and python3.

Platform support

The host selector lists Windows Server 2019, Server 2022, and Server 2025. Consult official Aqua system requirements documentation for the updated Windows support list and supported Linux distributions/versions. This prompt is informational, not an OS-version compatibility check.

Policy and credentials

Scope the Host Runtime Policy to the selected host and confirm connected Enforcer coverage in Aqua. Authentication tests require dedicated test-account credentials rather than silently reusing the administrator identity.

Exit behavior

Host-only option 0 exits without unrelated Kubernetes/Docker cleanup, but retries pending Windows registry cleanup. Main-menu option 47 performs broader cleanup.

08 / HOST RUNTIME

Host controls and authentication

Main menu	Controls
32-37	Cryptocurrency Mining; Brute Force Login Detection; DNS/IP Reputation; File Block; File Integrity Monitoring; Forensics
38-43	OS Users and Groups Allowed; OS Users and Groups Blocked; Package Block; Port Scanning Detection; Secure AI - Discovery; Secure AI - Unauthorized Models
44-45	Windows Registry Monitoring; Windows Registry Protection

Brute Force Login Detection

Use a disposable account and a Host Runtime Policy. The default is five attempts, with a maximum of ten. For password tests, intentionally enter an incorrect password through SSH. An optional key selects public-key authentication. A valid authorized key succeeds and stops the test, so it does not generate failures.

OS Users and Groups Allowed / Blocked

Use valid credentials. First establish a successful baseline login and verify identity/group membership. Then apply the allow/block policy and confirm one fresh SSH login. The workflow supports Linux or Windows with the matching Enforcer. It does not create users/groups or test RDP/console sign-in.

Disposable host activity

Host FIM creates, modifies, changes permissions on, and deletes files under an owned /tmp/aqua-probe-host-fim-* directory. It never edits real /etc/sudoers. Port scanning targets an explicitly authorized destination, with at most 16 concurrent one-second probes across ports 1-1024.

Evidence and recovery

Confirm host authentication logs and matching Aqua events. Keep recovery access, restore policy entries manually, and inspect printed owned paths if cleanup is interrupted. Bypass Scope and host System Integrity Monitoring have no dedicated host test functions.

09 / WINDOWS ONLY

Windows registry workflows

Both tests require **Windows Aqua Enforcer** and a Host Runtime Policy. Linux Enforcer cannot validate these signatures. The SSH preflight confirms OS/account identity, not Aqua installation or policy health.

1. Establish the disposable baseline

The workflow creates a unique, marked key at HKCU:\Software\AquaProbeLab-<random-id>. It prints the full policy path under HKEY_USERS\<SSH-account-SID> and its Operations subkey. Use the printed path and your tenant's descendant-matching syntax, not an existing OS/application registry path.

2. Monitoring: main menu 44

Enable **Create, Read, Modify, Delete, Attributes**, add the printed path and descendants to Monitored Paths, and sync the policy. Monitoring is audit only. Keep Protection off for the test path. Operations run separately over SSH. The ACL change is an attributes candidate whose event classification must be checked in Aqua.

3. Protection: main menu 45

Create the baseline while unprotected. Add the printed path and descendants to Protected registry paths in an enforcing policy, then sync before the create/modify/delete attempts. Review Aqua evidence. An ACL error or successful command alone does not prove protection behavior.

4. Remove protection and clean up

Remove only the test path from protection, wait for policy sync, then confirm cleanup. The script checks the account SID, random path, and ownership marker before deletion. Failed cleanup retains session state and blocks new Windows tests and termination until resolved.

Explicit cleanup

Use main option 46 or host-only option 16 to retry. After forced termination or lost state, inspect and remove only the printed owned key manually after removing protection. Audit records remain.

10 / AI VALIDATION

Secure AI workflows

Container application: main menu 27-29

The sample application uses **ericgomes56/ai-app:1.0**. Supply your own OpenAI API key at the hidden prompt. The deployment receives it through a Kubernetes Secret. Restrict access to the Secret and LoadBalancer endpoint, and never commit keys.

Open the exact URL printed from the created Service using **HTTP on port 8501**. If the LoadBalancer is pending, inspect mars-banking-app-lb in aqua-probe-lab and use its assigned hostname/IP. Subsequent Unauthorized Models and Protection tests can reuse the existing app until cleanup.

Test	Policy and application steps
Discovery	Enable discovery for the environment. In the app, open Request Simulator at the bottom right, choose OpenAI, and send a request. Review AI Findings and relevant audit events.
Unauthorized Models	Select OpenAI as Service and gpt-4 as Model Name (OpenAI@gpt-4). In Request Simulator, choose OpenAI and model override gpt-4, then send a request.
Protection	Aqua SaaS tenants only, not on-prem. Enable Prompt Protection Settings under Administration > Integrations > Secure AI. Open Attack Library and choose an example.

Host AI: main menu 42-43

Host tests send one native Python 3 HTTPS request over SSH, without deploying the sample container app. Enter a fresh key for each request. Match the selected available model to the policy. Confirm tenant/Enforcer feature coverage and model access.

Result interpretation

Requests may incur provider charges. API authentication, billing, connectivity, and unavailable-model errors do not establish an Aqua finding or denial.

11 / REFERENCE

CLI and image configuration

Option	Behavior
--help / -h	Print help and exit before Kubernetes checks.
--host-runtime	Use the host-only interface without Kubernetes startup checks.
--image / -i	Default test image: ericgomes56/aqua-probe:2.0. Use a pullable custom reference where supported.
--namespace / -ns	Aqua Enforcer namespace, default aqua. This does not change the test namespace aqua-probe-lab.
--daemonset / -ds	Custom Enforcer DaemonSet name. Built-in names are checked first.
--no-instructions / -n	Skip supported container prerequisites only. Host and other guarded workflows still require confirmation.
--version / -v	Print the embedded build date, then continue startup. This is not an exit-only command.
NO_COLOR=1	Disable colors. Redirected output and TERM=dumb also omit color codes.

Image exceptions and architecture

The non-compliant-image test uses **jerbi/eicar:latest**. Secure AI uses **ericgomes56/ai-app:1.0**. Other deployment prompts can use custom images. Generated workloads specify imagePullPolicy: Always.

Build images for the Kubernetes worker architecture, not merely the operator's Mac or Linux workstation. Read Dockerfile/README.md for AMD64/ARM64 paths and security caveats. The current build uses legacy CentOS/Go and disables some certificate checks.

12 / SESSION CLOSE

Cleanup and troubleshooting

Main menu: 47. Terminate Program

Termination retries pending Windows registry cleanup, then removes known Kubernetes test artifacts, named cluster roles/bindings, and the entire aqua-probe-lab namespace. It also removes the local Docker test container when Docker is available. Some deletion is asynchronous, so verify completion.

Run Docker-only tests on the intended worker with access to its Docker runtime. A container started elsewhere needs cleanup on that runtime. Neither Kubernetes access nor the namespace contains all host-level effects.

Symptom	Check before attributing it to Aqua
Container not found	Current pod and container names, namespace, rollout status, and container state.
Image will not start	Registry credentials, pullability, image architecture, admission decisions, and required commands.
No expected incident	Correct policy scope/mode, sync, feature coverage, trigger conditions, and event time filters.
SSH failure	Host key, route, port, credentials, OpenSSH configuration, and account lockout.
Cleanup blocked	Remove protection from only the owned test path, restore access, and retry. Inspect leftovers after interruption.

Close the workshop

Restore policy changes manually. Confirm lab resources and owned host/registry files are gone. Keep sanitized evidence and audit records. Do not assume every successful command produces an incident or that every failure is enforcement.

13 / RESOURCES

References and maintenance

Current project documentation

Repository and quick start

Control reference and detailed host instructions

Image build and architecture guide

Current terminal walkthrough

Source script

Offline test suites

Validation and support

Offline tests use mocks and do not establish live Aqua enforcement. Validate the selected control in a monitored, authorized lab. Report issues with the control name, platform, policy mode, expected outcome, and sanitized output. Use your organization's normal Aqua support channel for platform issues.

Guide maintenance

This guide reflects the repository reviewed on November 2, 2026. Menu numbers and platform coverage may change. Follow current script prompts and official Aqua system requirements for your deployment. The reproducible PDF source and supplied logo assets live in user-guide/.

Project attribution

Aqua Probe is a maintained fork and evolution of Aqua Warden. The project has been rebranded and is actively developed to support Aqua runtime security testing, PoVs, and technical enablement.

Tool developed by Eric Gomes, Senior DevSecOps Architect @ Aqua Security. No repository license file is currently included. Clarify licensing with the maintainer before reuse or redistribution.